

100%

Excellent!

Name: _____

These questions assess your understanding of the material from Chapter(s) 18, 19 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Suppose 66. mm separates two 1.7-m^2 plates in a parallel-plate capacitor. What charge is stored in this capacitor if it is connected to a 0.61-V battery?

ANSWER: $1.391 \times 10^{-10} \text{ C}$

$$\frac{Q}{V} = \epsilon_0 \frac{A}{d} \rightarrow Q = \frac{\epsilon_0 A V}{d}$$

$$Q = \frac{(8.85 \times 10^{-12} \frac{\text{C}}{\text{V}\cdot\text{m}})(1.7 \text{ m}^2)(0.61 \text{ V})}{0.066 \text{ m}} = \boxed{1.391 \times 10^{-10} \text{ C}}$$

2. A circuit has two branches: one branch has capacitors 1 and 2 while the second branch has capacitors 3 and 4. Capacitors 1 and 2 are connected in series and capacitors 3 and 4 are connected in series. The two branches are connected in parallel. Draw this circuit and find the total capacitance given that $C_1 = 2.7 \text{ nF}$, $C_2 = 12 \text{ nF}$, $C_3 = 8.9 \text{ nF}$, and $C_4 = 22 \text{ nF}$. Note: you must draw the circuit correctly to receive credit.

ANSWER: 8.541 nF

$$C_{12} = C_s = (2.7^{-1} + 12^{-1})^{-1} \text{ nF}$$

$$C_{34} = C_s = (8.9^{-1} + 22^{-1})^{-1} \text{ nF}$$

$$C_T = C_p = C_{12} + C_{34} = (2.7^{-1} + 12^{-1})^{-1} \text{ nF} + (8.9^{-1} + 22^{-1})^{-1} \text{ nF} = \boxed{8.541 \text{ nF}}$$

3. Suppose a 8.4-V custom battery can move $3.4 \times 10^3 \text{ C}$ of charge and a 12-V car battery can move $6.2 \times 10^4 \text{ C}$ of charge. What is the ratio of work done (or energy delivered) from the car battery compared to the custom battery?

ANSWER: $26.050 \text{ times more energy}$

$$\Delta V = \frac{\Delta PE}{q} \rightarrow \Delta PE = \Delta V q$$

$$\frac{\Delta PE_c}{\Delta PE_B} = \frac{\Delta V_c q_c}{\Delta V_B q_B} = \frac{(12 \frac{\text{J}}{\text{C}})(6.2 \times 10^4 \text{ C})}{(8.4 \frac{\text{J}}{\text{C}})(3.4 \times 10^3 \text{ C})} = \boxed{26.050}$$

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4. A heart defibrillator delivers 4.9×10^3 J of energy by discharging a capacitor initially at 4.7×10^4 V. What is its capacitance?

ANSWER: 4.436×10^{-6} F

$$E_{cap} = \frac{CU^2}{2} \rightarrow C = \frac{2E_{cap}}{U^2} = \frac{2(4.9 \times 10^3 \text{ J})}{(4.7 \times 10^4 \frac{\text{J}}{\text{C}})^2} = \boxed{4.436 \times 10^{-6} \text{ F}}$$

$$\frac{C^2}{\text{J}} = \frac{C}{\frac{\text{J}}{C}} = \frac{C^2}{\text{J}}$$

5. Calculate the magnitude of the force exerted on a 0.38-nC charged particle placed in a $4.7 \times 10^5 \frac{\text{N}}{\text{C}}$ electric field.

ANSWER: 1.786×10^{-4} N

$$E = \frac{F}{q} \rightarrow F = Eq = (4.7 \times 10^5 \frac{\text{N}}{\text{C}})(0.38 \times 10^{-9} \text{ C}) = \boxed{1.786 \times 10^{-4} \text{ N}}$$

6. Calculate the final speed of a $2.9 \times 10^{-18}\text{-C}$ charged particle that accelerates from rest through a potential difference of 1.1 V. The mass of the charged particle is 8.4×10^{-28} kg.

ANSWER: $8.715 \times 10^4 \frac{\text{m}}{\text{s}}$

$$\Delta E_t = 0 = \Delta KE + \Delta PE = KE_f - KE_i + PE_f - PE_i \rightarrow KE_f = PE_i = \frac{1}{2}mv^2 = \Delta Uq$$

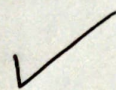
$$v = \sqrt{\frac{\Delta Uq}{\frac{1}{2}m}} = \sqrt{\frac{(1.1 \frac{\text{J}}{\text{C}})(2.9 \times 10^{-18} \text{ C})}{\frac{1}{2}(8.4 \times 10^{-28} \text{ kg})}} = \boxed{87150.663 \frac{\text{m}}{\text{s}}}$$

$$\frac{\frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}}{\text{kg}}$$

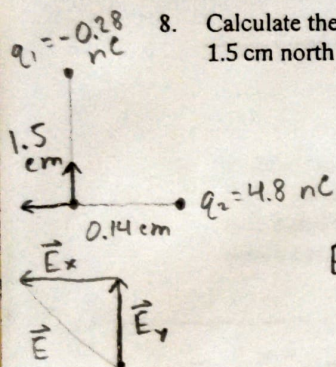
7. A particular Van de Graaff generator has a metal sphere of diameter 60. cm that produces a voltage of 81.6 kV near its surface. What excess charge resides on the sphere?

$$V = \frac{kQ}{r} \rightarrow Q = \frac{Vr}{k} = \frac{(81600 \frac{\text{N}\cdot\text{m}}{\text{C}})(0.3 \text{ m})}{8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}} = \boxed{2.723 \times 10^{-6} \text{ C}}$$

ANSWER: $2.723 \times 10^{-6} \text{ C}$



8. Calculate the magnitude of an electric field measured at the origin due to the presence of two point charges q_1 , q_2 . q_1 is located 1.5 cm north of the origin and q_2 is located 0.14 cm east of the origin. $q_1 = -0.28 \text{ nC}$ and $q_2 = +4.8 \text{ nC}$. $\frac{\text{N}}{\text{C}}$

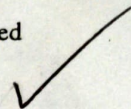


$$E = \sqrt{E_x^2 + E_y^2} = \sqrt{\left(\frac{kQ_2}{r_2^2}\right)^2 + \left(\frac{kQ_1}{r_1^2}\right)^2}$$

$$E = \sqrt{\left(\frac{(8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2})(4.8 \times 10^{-9} \text{ C})}{(0.0014 \text{ m})^2}\right)^2 + \left(\frac{(8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2})(0.28 \times 10^{-9} \text{ C})}{(0.015 \text{ m})^2}\right)^2}$$

$$E = \boxed{22016329.37 \frac{\text{N}}{\text{C}}}$$

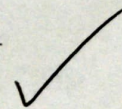
ANSWER: $2.202 \times 10^7 \frac{\text{N}}{\text{C}}$



9. What is the electric potential 0.79 cm away from the center of a metal sphere that has a 2.2-nC static charge?

$$V = \frac{kQ}{r} = \frac{(8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2})(2.2 \times 10^{-9} \text{ C})}{0.0079 \text{ m}} = \boxed{2503.544 \text{ V}}$$

ANSWER: 2503.544 V



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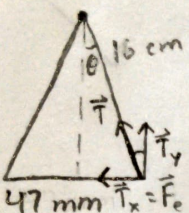
10. Calculate the strength of an electric field created by a 0.26-nC charged particle at a distance of 8.2 mm away.

ANSWER: $3.476 \times 10^4 \text{ N/C}$

$$E = \frac{k|Q|}{r^2} = \frac{(8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2})(0.26 \times 10^{-9} \text{ C})}{(0.0082 \text{ m})^2} = 34762.046 \frac{\text{N}}{\text{C}}$$

11. Two identical charged spheres, each with a mass of 26 g , hang in equilibrium. They are suspended by a string of length 16 cm and are separated by a horizontal distance of 47 mm . Calculate the magnitude of the charge on each sphere.

ANSWER: $9.642 \times 10^{-8} \text{ C}$



$$\sin \theta = \frac{0.0235 \text{ m}}{0.16 \text{ m}} \rightarrow \theta = \sin^{-1}\left(\frac{0.0235 \text{ m}}{0.16 \text{ m}}\right) = 8.446^\circ$$

$$\cos \theta = \frac{T_y}{T} = \frac{mg}{T} \rightarrow T = \frac{mg}{\cos \theta}$$

$$\sin \theta = \frac{T_x}{T} = \frac{\frac{kq^2}{r^2}}{\frac{mg}{\cos \theta}} = \frac{kq^2 \cos \theta}{r^2 mg} \rightarrow q = \sqrt{\frac{\sin \theta r^2 mg}{k \cos \theta}} = \sqrt{\frac{\tan \theta r^2 mg}{k}}$$

$$q = \sqrt{\frac{\tan(8.446^\circ)(0.047 \text{ m})^2(0.026 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})}{8.99 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}}} = 9.642 \times 10^{-8} \text{ C}$$

12. Dry air will support electric fields up to its dielectric strength, $3.0 \times 10^6 \frac{\text{V}}{\text{m}}$. Above that value, an electric field creates enough ionization in the air to make the air a conductor, which causes a discharge or spark to reduce the field. Calculate the maximum voltage between a charged region at the base of a thundercloud and the ground if they are separated by 1.2 km of dry air. Model the cloud and ground as two parallel conducting plates.

ANSWER: $3.6 \times 10^9 \text{ V}$

$$\Delta V = E d = (3 \times 10^6 \frac{\text{V}}{\text{m}})(1200 \text{ m}) = 3.6 \times 10^9 \text{ V}$$

1. $Q = 1.39 \times 10^{-10} \text{ C} \therefore C = \frac{Q}{V} = \epsilon_0 \frac{A}{d}$
(Chapter 19, Section (5) Capacitors and Dielectrics - cple_ch19_ex8)
2. $C_T = 8.54 \times 10^{-9} \text{ F} \therefore \frac{1}{C_s} = \frac{1}{C_1} + \dots + \frac{1}{C_n}, C_p = C_1 + \dots + C_n$
(Chapter 19, Section (6) Capacitors in Series and Parallel - cple_ch19_ex10)
3. $W_{ratio} = 26.1 \text{ times more energy} \therefore \Delta V = \frac{\Delta PE}{q}, W = -\Delta PE \text{ for a conservative force, so } W = -q\Delta V$
(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cple_ch19_ex1)
4. $C = 4.44 \times 10^{-6} \text{ F} \therefore E_{cap} = \frac{CV^2}{2}$
(Chapter 19, Section (7) Energy Stored in Capacitors - cple_ch19_ex11)
5. $F = 1.79 \times 10^{-4} \text{ N} \therefore F = Eq$
(Chapter 18, Section (4) Electric Field - cple_ch18_ex3)
6. $v_f = 8.72 \times 10^4 \frac{\text{m}}{\text{s}} \therefore \Delta V = \frac{\Delta PE}{q}, W_{c.f.} = -\Delta PE \therefore W = -q\Delta V, W_{net} = \Delta KE \therefore \Delta KE = -q\Delta V$
(Chapter 19, Section (1) Electric Potential Energy: Potential Difference - cple_ch19_ex3)
7. $Q = 2.72 \times 10^{-6} \text{ C} \therefore V = \frac{kQ}{r}$
(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cple_ch19_ex7)
8. $E = 2.20 \times 10^7 \frac{\text{N}}{\text{C}} \therefore E = k \frac{|Q|}{r^2}, E^2 = E_x^2 + E_y^2$
(Chapter 18, Section (5) Electric Field Lines - cple_ch18_ex4)
9. $\Delta V = 2.50 \times 10^3 \text{ V} \therefore V = \frac{kQ}{r}$
(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cple_ch19_ex6)
10. $E = 3.48 \times 10^4 \text{ N/C} \therefore E = k \frac{|Q|}{r^2}$
(Chapter 18, Section (4) Electric Field - cple_ch18_ex2)
11. $q = 9.64 \times 10^{-8} \text{ C} \therefore \Sigma \vec{F}_x = m\vec{d}_x, F = k \frac{|q_1 q_2|}{r^2}$
(Chapter 18, Section (3) Coulomb's Law - PSE_Serway_4e_ch23_ex5)
12. $\Delta V_{max} = 3.60 \times 10^9 \text{ V} \therefore \Delta V = Ed$
(Chapter 19, Section (2) Electric Potential in a Uniform Electric Field - cple_ch19_ex4)